# INDIVIDUAL WEEKLY PROJECT REPORTS

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**Role: Exploratory Data Analysis and Data Visualization**
**Project Title: Flight Delay Analysis and Visualization**

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## WEEK 6 – Advanced Visualization and Feature-Based Comparisons

During the sixth week, I continued exploring the dataset and focused on preparing visual comparisons using the relevant features identified by the team. Feature preparation helped organize the information into useful categories that could make visual analysis easier.

I explored how flight-related variables could be grouped to study differences across airlines, airports, and time periods. I understood that categorizing information could help reveal patterns that might otherwise be difficult to identify from individual flight records.

I also considered the importance of comparing different numerical variables through suitable graphs. For example, scatter plots could

help examine relationships, while bar charts could summarize differences between categories.

Another part of my work involved reviewing the visualizations prepared during the earlier stages and considering whether they provided useful information. I focused on avoiding unnecessary charts and selecting visualizations that directly supported the project objectives.

I coordinated with the preprocessing team to understand the variables that were available after data cleaning and feature preparation. This helped ensure that the visual analysis used appropriate data.

**Outcome and Learning:**

This week helped me understand how feature engineering and visualization work together. I learned that organizing data into meaningful categories can improve comparisons and make analytical findings easier to interpret.